

Element F: Consideration of Design Viability

Student E, a student in our engineering class, needs a way to get rid of wasps, hornets, and other insects that enter his room at night when he sleeps. The student's bedroom has a serious insect problem. The insects make the space noisy and affect the students' ability to rest.

Our design will lure in the insects with a light at the back of the trap that we are confident will be an effective bait because of the information our stakeholders have provided about the insects. Student E stated in the initial problem definition phase that the insects would swarm around his phone in the dark room at night. This information was crucial in the development of our design, as it informed us that the insects were attracted to light. The design will trap insects that have been lured in using insect-catching sticky paper, which is a common solution to this type of problem, and which can be easily purchased online.

Our prototype is very simple to design as it only requires three parts. The lure for the insects will be a flashlight that is placed at the end of the trap; the body of the trap will be 3D-printed, and the sticky paper will be purchased online. These materials are easily obtainable, and the prototype is simple to construct. One feature our design has that is not part of the design requirements is the ability to be easily reusable. Our design consists of three parts: a main cylinder, a flashlight fixed to the back, and a cylinder of sticky paper lining the main cylinder. The main body and flashlight will not need to be replaced, and the sticky paper is easily removable and changeable, which makes it easy to be reused repeatedly.

The sticky paper idea is similar to an existing pest-control product known as a "sticky trap", a double-sided piece of paper that is coated with a sticky residue to catch insects that land on it. The sticky paper in our design, however, is only one-sided in order not to stick to the main body of the prototype.